

6 Nucleic acids and protein synthesis

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

DNA is antiparallel, with A-T and C-G complementary pairing. Semi-conservative replication yields one old and one newly synthesised strand per molecule. Transcription produces RNA from a DNA template; translation reads mRNA codons at ribosomes while tRNA brings amino acids. A substitution may be silent, missense or nonsense depending on codon and reading frame.

Worked example: follow the reasoning

If template DNA triplet is TAC, complementary mRNA is AUG, coding for methionine in the standard code.

Quick check · multiple choice

A Which molecule carries codons read by ribosomes?

[1]

A DNA polymerase

B mRNA

C ATP synthase

D Haemoglobin

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A DNA template segment 3'-TAC GAA-5' is transcribed. Give mRNA 5'-3' and explain why changing one base does not always change protein.

[3]

Answers and explanation

Quick check: B. mRNA

Question 1 [3 marks]

mRNA 5'-AUG CUU-3' [2]. Genetic code is degenerate: a changed codon can specify the same amino acid [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.